Mock Interview Edition

LLM Inference Serving at Scale

17 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

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HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

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Every scenario, by part

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03 Memory or compute: what limits the batch? Mechanism 7

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SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

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02 Review the fleet-size estimate Sizing 5
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14 What availability can we promise? Availability 23
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SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the problem: targets, memory and cost

Turning a vague request into numbers, then checking the numbers against what the hardware can do.

01 Write the targets before the architecture

FRAMING

SCENARIO

A product manager asks for chat serving for a 13B-parameter model that is “fast and cheap.” Engineers are already arguing about the batch size. About 1,000 conversations will be active at once, with 2,000 tokens of context on average and 500 generated tokens each.

OPENING PROBE

What targets do you write down?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Which two targets pull against each other?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Can a 200 ms p50 TTFT be met for a 2,000-token prompt on one GPU?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How do you write the TTFT target so it can be met honestly?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 6

02 Review the fleet-size estimate

SIZING

SCENARIO

A colleague sizes the fleet for 1,000 concurrent conversations on 80 GB GPUs serving a 13B-parameter model in 16-bit precision. Their notes are below, and they conclude that 24 GPUs are enough. Find what is wrong.

```
# the colleague's estimate
kv_per_token = 2 * 32 * 4096 * 2          # layers, hidden size, bytes -> 512 KB
request_gb   = 1.22                       # 2,500 tokens
per_gpu      = 52 / 1.22                  # 42 requests per GPU
gpus         = 1000 / 42                  # 24 GPUs
```

OPENING PROBE

What is wrong with the per-token figure?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

Redo the chain with the right figure.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What else does the estimate assume?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

Is sizing every request at its peak length right?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What do you verify before ordering hardware?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

03 Memory or compute: what limits the batch?

MECHANISM

SCENARIO

The team notices that adding requests to the batch raises throughput until it stops helping. Someone proposes buying GPUs with more compute. Another says memory is the problem. You have the sizes from Question 2 and no benchmarks.

OPENING PROBE

What does one decode step have to read?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

How long does that take?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

Why does batching raise throughput at all?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What stops you adding more requests?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What would you buy, and what would you change?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

04 Price a million tokens

COST

SCENARIO

Finance asks for the cost of a million generated tokens and whether the memory-management technique matters to the bill. Use a GPU price of $2.50 an hour (illustrative) and the sizing from Question 2.

OPENING PROBE

What is the cost per million tokens for the block-allocated fleet?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does the memory technique change?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What does the calculation assume that real traffic will not satisfy?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

Where is the next saving?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

How do you track it in production?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 10

PART 2

PagedAttention: from operating-system paging to KV blocks

Why contiguous allocation wastes most of the cache, how blocks and block tables fix it, and what sharing blocks makes possible.

05 Why contiguous allocation wastes 60 to 80%

MECHANISM

SCENARIO

A teammate argues: “Memory is cheap. Give each request one contiguous region sized for the maximum length, 2,048 tokens, and keep the code simple.” You serve the 13B model, 0.8 MB of KV cache per token, and a typical conversation ends at about 400 tokens.

OPENING PROBE

Where does the waste come from?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What did the paper measure?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why not just compact memory now and then?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does paging fix, and what does it leave?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

06 Find the bug in the block allocator

CODE REVIEW

SCENARIO

A new engineer has written the allocator below. It passes a simple test where requests start with an empty prompt. In staging, long prompts produce garbage output, and sometimes the process crashes under load. Block size is 16 tokens.

```
BLOCK_SIZE = 16

class Allocator:
    def __init__(self, total_blocks):
        self.free = list(range(total_blocks))
        self.tables = {}

    def on_decode_step(self, request_id, generated):
        # called once per decode step with the number of tokens generated so far
        table = self.tables.setdefault(request_id, [])
        if generated > len(table) * BLOCK_SIZE:
            table.append(self.free.pop())
```

OPENING PROBE

What does it do wrong for a request with a 40-token prompt?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What goes wrong when memory is full?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

State the corrected rule.

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you test?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

07 Choose the block size

DESIGN CHOICE

SCENARIO

Someone proposes 256-token blocks because they make the attention kernel more efficient. Another proposes 4-token blocks to waste almost nothing. The default is 16. Requests average 2,500 tokens, with many short ones in the mix.

OPENING PROBE

What does the block size trade?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How much space does each choice waste?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Whose workload changes the answer?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you measure?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 15

08 Twenty users share a system prompt

SHARING

SCENARIO

An assistant product prepends the same 1,000-token system prompt to every conversation. Twenty conversations are running on one GPU. An engineer asks whether the prompt's KV cache can be stored once instead of twenty times.

OPENING PROBE

How much memory does sharing save?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How does a request write to a shared block safely?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What stops two prompts from sharing?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does sharing do to the first-token wait?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

PART 3

Scheduling, caching and routing

What the scheduler does when memory runs out, how a long prompt hurts everyone, and where cached prefixes live.

09 The GPU is out of blocks in the middle of decoding

OPERATIONS

SCENARIO

At the evening peak, KV-cache utilization hits 100%. Every running request needs one more block on the next step and the free list is empty. Right now the process throws an exception and the whole batch fails.

OPENING PROBE

What are the options when no block is free?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Whom do you preempt?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Swap or recompute? Do the arithmetic.

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How do you know it is happening, and what do you do about it?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

10 One long prompt stalls every stream

DEBUGGING

SCENARIO

p99 inter-token latency spikes to about 800 ms a few times an hour, although average utilization is moderate. The logs show that each spike coincides with a request whose prompt is around 8,000 tokens. The target is 120 ms.

OPENING PROBE

Why would one request hurt all the others?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What is the fix?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

How do you choose the chunk size?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

How do you confirm the diagnosis?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What other options exist?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

11 Keep the conversation's cache, or recompute it?

CACHE DESIGN

SCENARIO

Users in a chat product reply about 60 seconds after each answer. The context is 2,000 tokens. The team wants first-token times near 200 ms and asks whether to keep each conversation's KV cache on the GPU between turns.

OPENING PROBE

What does a cache hit save?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does it cost to park the cache on the GPU while the user thinks?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

So where should an idle cache live?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What hit rate does the target need?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What did Character.AI say about this?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

12 The load balancer is round-robin

ROUTING

SCENARIO

Eight replicas sit behind a round-robin load balancer. The prefix-cache hit rate is 12%, first-token times are poor, and chat users send several turns per conversation.

OPENING PROBE

Why is the hit rate near 12%?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What is the fix?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What does affinity cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What happens when you add a replica?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

How do you verify the change?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 20

PART 4

Operating at scale: dashboards, availability, growth and safety

Reading the signals, promising an availability number, planning for ten times the load and closing the side channels.

13 Three incidents, one dashboard

MONITORING

SCENARIO

In one week the dashboard shows three different problems. (A) KV utilization 99%, queue depth growing, p99 first-token time rising. (B) p99 first-token time breached while utilization is 55% and the queue is short. (C) Cost per million tokens up 30% with flat latency and flat traffic.

OPENING PROBE

What is A, and what do you do?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is B?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is C?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What do you put on the dashboard so these are distinguishable?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 22

14 What availability can we promise?

AVAILABILITY

SCENARIO

A gateway, an inference backend and a vector database sit in series. Their availabilities are 99.95%, 99.9% and 99.99%. Sales wants to promise 99.99% for the whole chain.

OPENING PROBE

What is the availability of the chain?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

Where does redundancy pay most?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

Can you reach 99.99%?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What does the formula leave out?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What do you promise?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 23

15 Traffic grows tenfold

SCALE

SCENARIO

The product succeeds. Concurrent conversations are expected to grow from 1,000 to 10,000 in a year. Your manager asks what it costs and what breaks first.

OPENING PROBE

What does ten times do to the plain plan?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Which levers shrink the fleet?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What about demand spikes?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What changes in operations at 400 GPUs?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 24

16 Two tenants share a prefix cache

SECURITY

SCENARIO

The API is multi-tenant. Automatic prefix caching is on because it made first-token times much better. A security reviewer asks what could go wrong.

OPENING PROBE

What is the risk?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the documentation offer?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does isolation cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What other abuse cases belong in the review?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 25

17 A blog post says 33 times cheaper

EVIDENCE

SCENARIO

An executive forwards a post from Character.AI: “reduced our serving costs by at least 33X.” They ask the team to “do what they did” and promise the same saving next year.

OPENING PROBE

What does the post actually say?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the 33 times measured against?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What can you borrow?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What do you tell the executive?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 26

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	Write the targets before the architecture	5			
02	Review the fleet-size estimate	6			
03	Memory or compute: what limits the batch?	6			
04	Price a million tokens	6			
05	Why contiguous allocation wastes 60 to 80%	5			
06	Find the bug in the block allocator	5			
07	Choose the block size	5			
08	Twenty users share a system prompt	5			
09	The GPU is out of blocks in the middle of decoding	5			
10	One long prompt stalls every stream	6			
11	Keep the conversation's cache, or recompute it?	6			
12	The load balancer is round-robin	6			
13	Three incidents, one dashboard	5			
14	What availability can we promise?	6			
15	Traffic grows tenfold	5			
16	Two tenants share a prefix cache	5			
17	A blog post says 33 times cheaper	5			